

SUMMARY AI engineer with hands-on industry experience building GraphRAG systems, agentic LLM workflows, and LLM-based document extraction pipelines. MSc in Data Science & AI (University of Florence, 2026). Research interests: explainable AI, automated fact-checking, and robustness of retrieval-augmented systems.

- EXPERIENCE**
- Futureforce Builder Lab Finalist** — Salesforce Dublin, Ireland · Sep 2026
- Reached finalist stage in Salesforce's AI builder assessment; assessed on prompt engineering, coding and deployment, and **AI tool** fluency.
 - Worked with an **AI coding agent** in a timed 90-minute build on an unfamiliar codebase, fixing a bug and shipping a working feature under live observation by two interviewers.
 - Presented and defended the solution in an executive pitch and technical review.
- AI & Data Science Intern** — Dometria Florence, Italy · October 2025 – Present
- Designed a **GraphRAG** asset–investor matching system on ColdRAG and DialKG to tackle cold-start matching.
 - Built an LLM extraction pipeline turning ~92 real-estate contracts into validated structured JSON with **Pydantic** schemas.
 - Designed a hybrid-retrieval query pipeline with cross-encoder reranking and MMR diversification for listing matching.
 - Migrated the internal conversational assistant to an event-driven **LlamaIndex** workflow with FAISS vector search.
 - Fine-tuned OpenAI **Whisper** on domain-specific real-estate jargon.
- Freelance AI Data Annotator** — Outlier AI Remote · Jan 2025 – Present
- Evaluated and annotated data for LLM training under strict quality guidelines.
- Academic Tutor** — University of Florence Florence, Italy · October 2025 – Present
- Tutored students in **Python** programming with running practical exercises and problem-solving sessions.
- Python Instructor** — Riverloop Messina, Italy · Nov 2024 – Dec 2024
- Taught Python for data analysis, ML, and deep learning; built the course GitHub repository with notebooks and exercises for 20 students.

- PROJECTS**
- MSc Thesis — From Documents to Matches: Evidence-Grounded Extraction and GraphRAG Cold-Start Recommendation for Real-Estate Off-Market Intermediation** — LLM document extraction and **GraphRAG** cold-start matching for real estate, in industry collaboration with Dometria. Python, LlamaIndex, knowledge graphs.
- GraphRAG-CLI** — github.com/vivri161803/graphrag-cli — dual-paradigm agentic RAG CLI with graph- and vector-based retrieval twins for side-by-side comparison. Python, **LangGraph**, FAISS.
- graphrag-probe** — github.com/vivri161803/graphrag-probe — multi-agent GraphRAG evaluation and red-teaming system with builder, adversary, and judge agents exposed via **MCP**.
- Explainable GNNs on Knowledge Graphs** — applied **GraphLIME** to RDF knowledge graphs in an XAI lab project. PyTorch Geometric, RDF.

- EDUCATION**
- MSc Data Science & AI** — University of Florence · 108/110 Sep 2024 – Oct 2026
- BSc Management Engineering** — University of Florence · 100/110 Sep 2021 – Jul 2024

SKILLS	Languages	Python (advanced), SQL, Java, Bash, LaTeX
	ML / DL	PyTorch, Keras, scikit-learn, Hugging Face Transformers, pandas, NumPy
	LLMs & Retrieval	LangGraph, LangChain, LlamaIndex, RAG, GraphRAG, knowledge graphs, FAISS, MCP, Pydantic
	Speech & Document AI	ASR, Whisper fine-tuning, torchaudio, OCR, VLMs
	Explainable AI	GraphLIME
	Infrastructure	Git, Docker, Linux, HPC clusters

- LANGUAGES & VOLUNTEERING**
- Languages** Italian (native), English (professional, C1)
- Volunteering** Volunteer, Misericordia (Prato) — First Aid and AED certified.